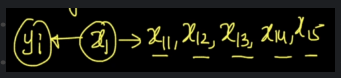
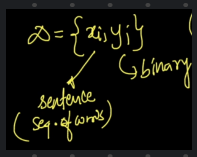
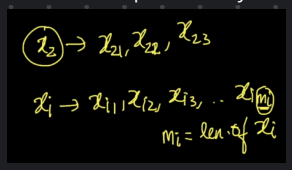
Recurrent Neural Networks

Consider the Amazon FoodReviews example (binary classification).

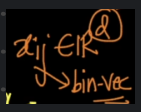




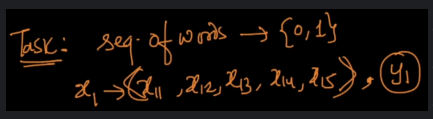
A sentence xi consists of multiple words.
Words are represented by one-hot encoding.



Say the size of vocab is d.
Then each word is a d-dim vector.



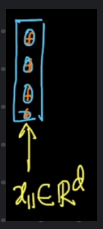
Task: classify a sequence of words to {0, 1}.



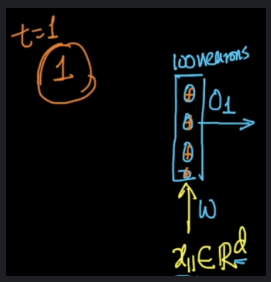
Task: classify a sequence of words to {0, 1}.

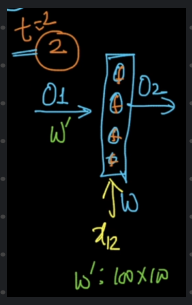
We'll process word by word. First all sentences in x1, then in x2, and so on.

Say 100 neurons.



Consider the first word as input to a hidden layer of neurons (activation function f).
We have a (d x 100) weight matrix.
We generate an output, say O1. O1 = f(w·x11)

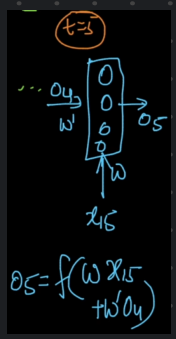


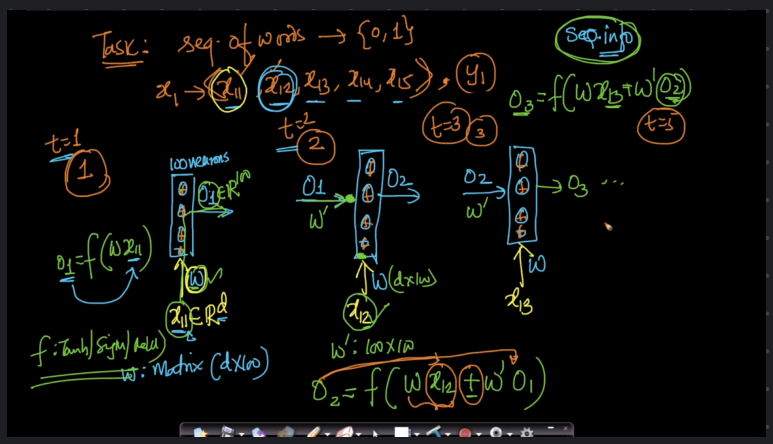


At the next timestep, we give O1 along with the next word to the layer.
This is done to preserve sequence information.

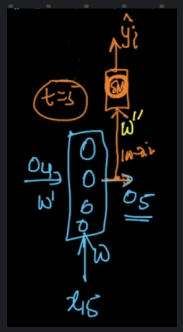
O2 = f(w·x12 + w'·O1)

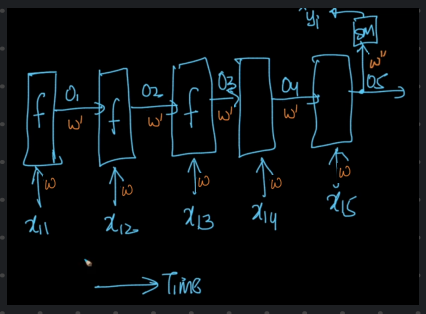
This will go on till t=5, since we've got 5 words.



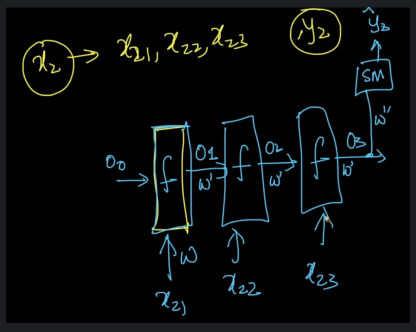


Now to generate y1, we take the 5th output O5, at the end put a softmax (or logistic) layer, to get a binary output.

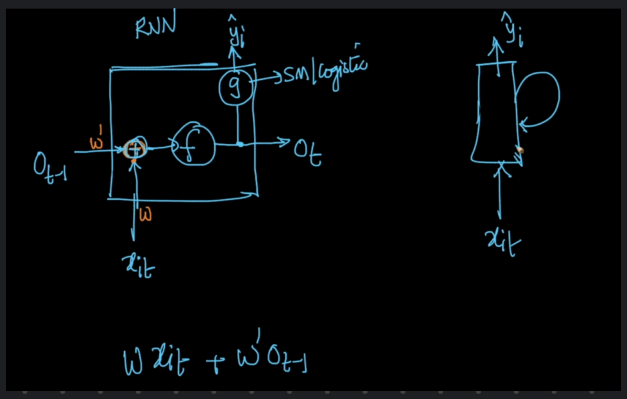




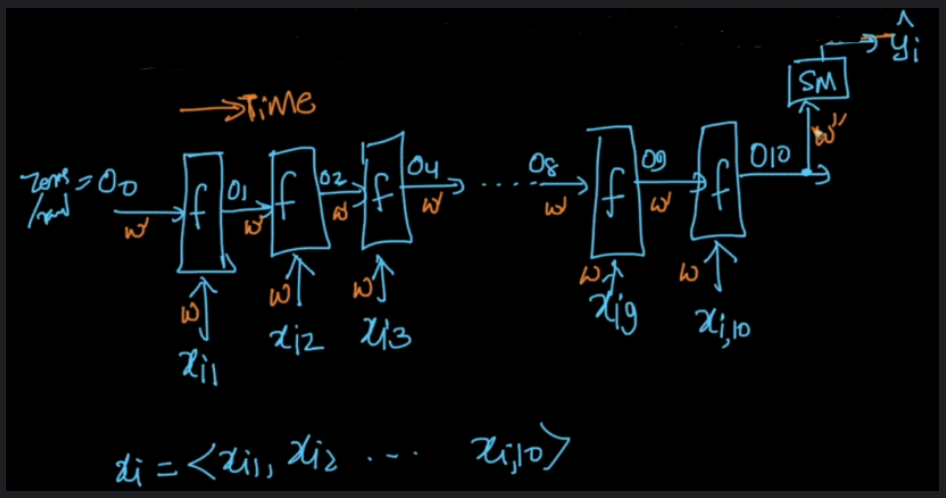
Now we're done with the first word. Same happens with the 2nd with a different # of time steps.



The same thing is represented as:



Backpropogation over time



First forward prop takes place.

$$O_1 = f(W x_{i1} + W' O_0)$$
$$O_2 = f(W x_{i2} + W' O_1)$$
$$\vdots$$
$$O_{10} = f(W x_{i10} + W' O_9)$$
$$\hat{y}_i = g(W'' \cdot O_{10})$$

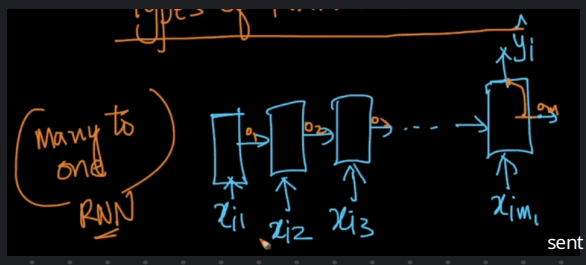
Backprop then takes place in the opposite direction of the arrows.

$$\frac{\partial L}{\partial \hat{y}_i}, \quad \frac{\partial L}{\partial O_{10}} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial O_{10}}$$

$$\frac{\partial L}{\partial W''} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial W''}$$

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial O_{10}} \cdot \frac{\partial O_{10}}{\partial W}$$

Problem: The weight modified at the last timestep is calculated by dL/dw which would be a product of a lot of partial derivatives. Vanishing gradient is observed.



sentiment analysis

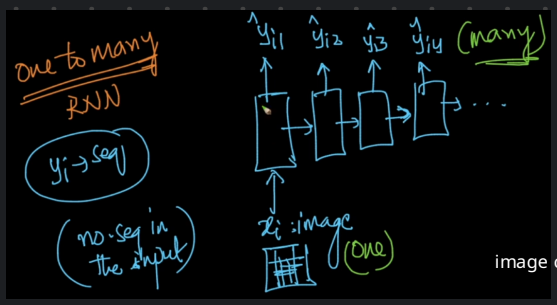
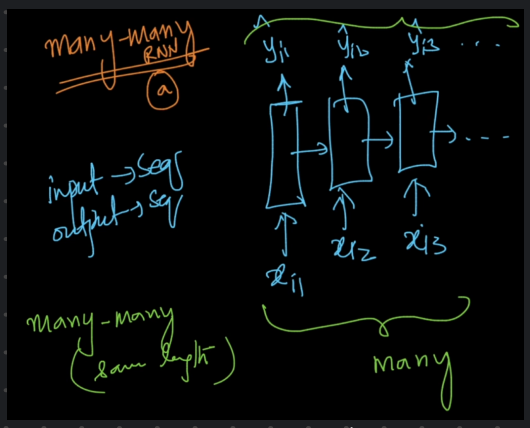
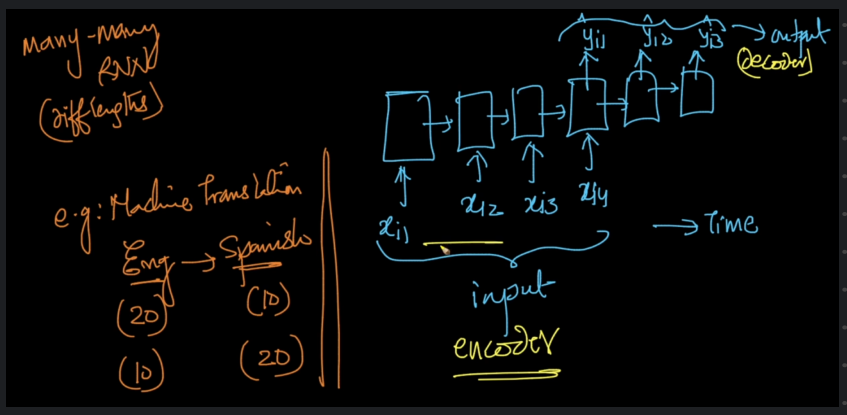
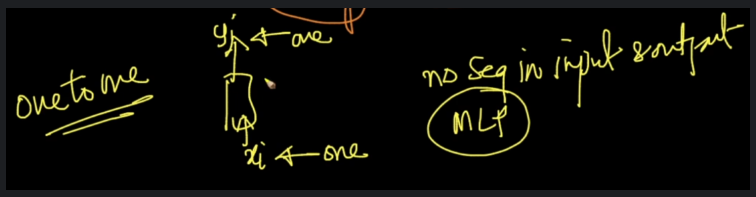


image captioning



pos tagging





Consider an alternative representation of RNNs.
Instead of adding xit and Ot-1, we concatenate/merge them (like an augmented matrix).

